

Ash Shati Ash Sharqi District

AI-powered geospatial analysis identifying critical infrastructure gaps, regulatory compliance issues, and investment opportunities across 19 GIS layers.

PREPARED FOR

Urban Planning & Lands
Agency Leadership ·
MOMAH

DATA SOURCE

EPM GIS Geodatabase ·
19 Layers

DISTRICT AREA

4.48 km² · 16,380
residents

Spatial Overview of the District

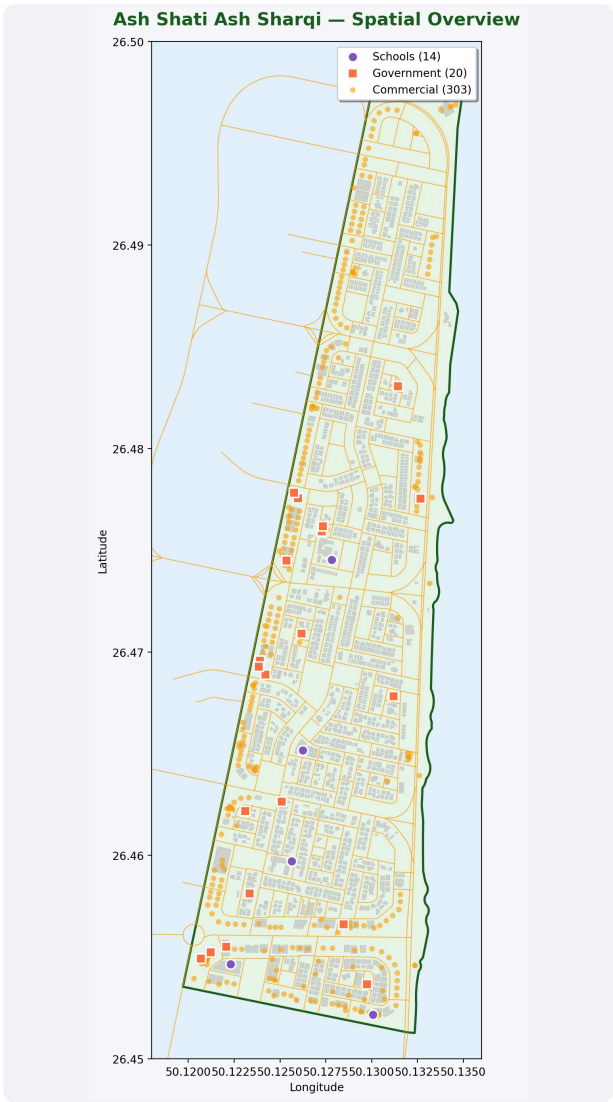
Ash Shati Ash Sharqi sits along the coastal frontage of Dammam, extending across **4.48 km²** in a long, narrow ribbon paralleling the Arabian Gulf. The aerial view shows the district's geographic position and its relationship to the surrounding urban fabric, while the analytical map illustrates the spatial distribution of educational, governmental, and commercial facilities derived from the underlying GIS layers.

Geographic Location & Surrounding Urban Fabric



Aerial imagery · District administrative boundary highlighted in blue

Spatial Distribution of Key Facilities



Analytical map · Schools (14) · Government (20) · Commercial (303)

Executive Summary

This report demonstrates the strategic value of deploying AI-powered GIS analysis. By cross-referencing 19 geospatial layers against urban planning benchmarks, the **Rashid** digital agent has instantly identified critical infrastructure gaps, regulatory compliance issues, and investment opportunities in Ash Shati Ash Sharqi.

10 Key Business Insights · Ash Shati Ash Sharqi District, Dammam

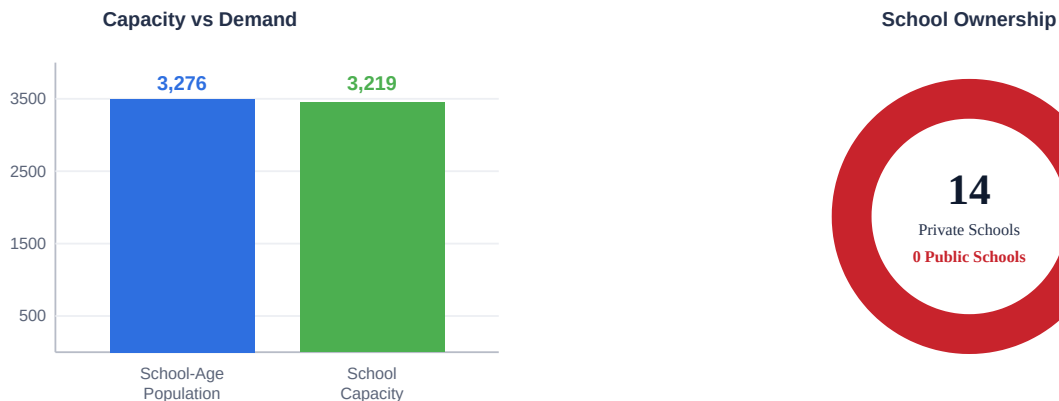
CRITICAL 0 Public Schools	CRITICAL 778 Unlicensed Developments	OPPORTUNITY 0 Health Centers	CRITICAL 2 Parks in 4.48 km²	HIGH 14.2/km Street Lights vs 30-40
MEDIUM 0.13 Parking per Commercial	MEDIUM 1 Jami Mosque (needs 2-3)	OPPORTUNITY 27.9 ha Undeveloped Land	POSITIVE 7.5 Buildings per Hydrant	POSITIVE 23 Mosques (1 per 712)

01 Diversity Gap in Educational Institutions

CRITICAL

The district has a total school capacity of **3,219**, which adequately covers **98.3%** of the estimated school-age population (3,276 children). However, all **14 schools are private (أهلية)**, with no government educational facilities registered in the district. This represents a **diversity gap in available educational options** for residents, warranting consideration of complementary government schools to broaden institutional variety and parental choice.

Insight 1 · All Schools Private — Government Option Absent

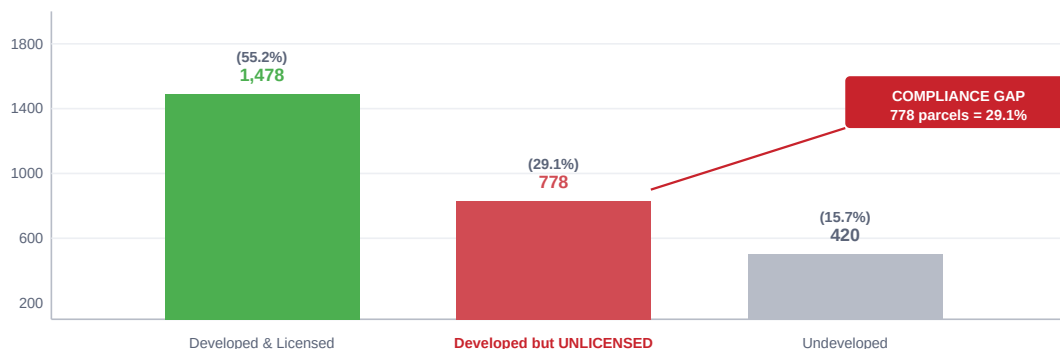


02 Regulatory Compliance · The "Unlicensed Development" Gap

CRITICAL

While 2,256 parcels (84.3%) are physically developed, the active building permits layer only shows 1,478 licensed parcels. This reveals a **compliance gap of 778 parcels (29.1%)** that are developed but lack active licenses in the municipal system. This represents a significant opportunity for regularization and municipal revenue generation.

Insight 2 · 778 Parcels Developed Without Active Licenses (29.1%)

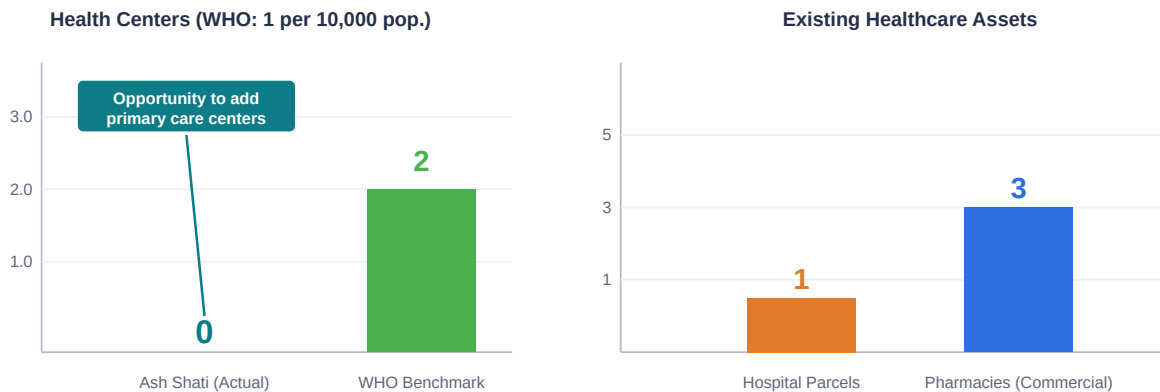


03 Opportunity to Enhance Primary Healthcare Capacity

OPPORTUNITY

For a population of 16,380 residents, the World Health Organization (WHO) benchmark requires at least **1.5 primary health centers** (1 per 10,000 residents). GIS data indicates an **opportunity to strengthen primary healthcare capacity** in the district, supported by existing infrastructure that includes one parcel zoned for a general hospital and three commercial pharmacies — a foundation that can be built upon to complete the primary care network.

Insight 3 · Opportunity to Strengthen the Primary Care Network



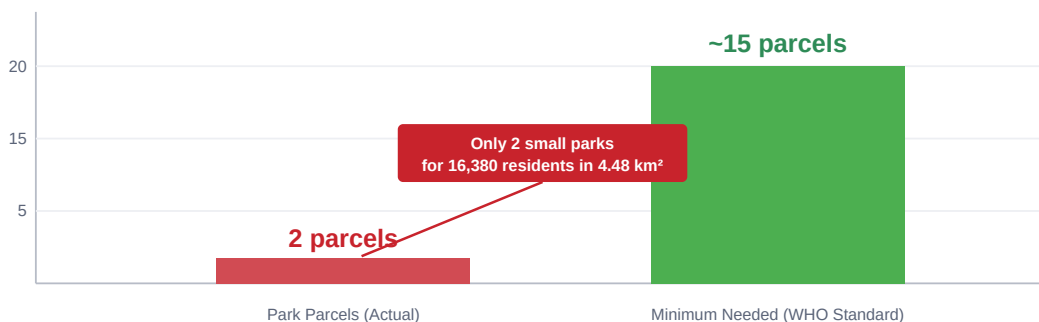
04 Severe Green Space Deficit

CRITICAL

The WHO recommends a minimum of **9 m² of green space per capita**. For this district, that equates to a requirement of **147,420 m²**. The GIS data identifies only **2 small parcels** designated as parks or recreational spaces across the entire 4.48 km² district, falling drastically short of urban quality-of-life standards.

Insight 4 · Severe Green Space Deficit

WHO requires 9 m² per capita = 147,420 m² needed



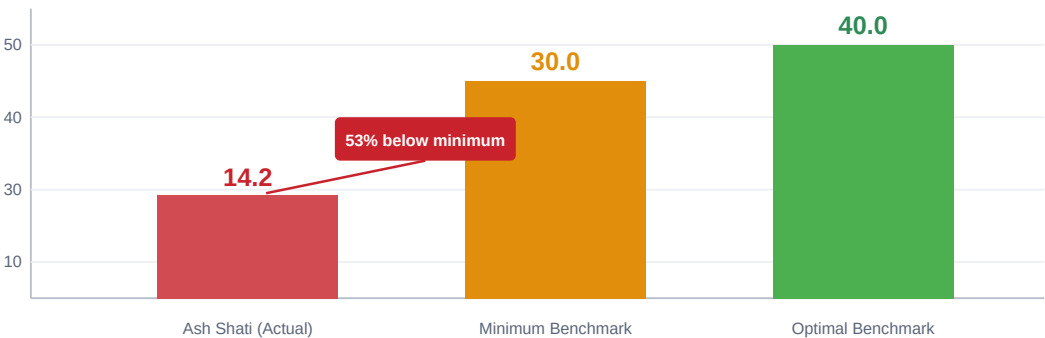
05 Street Lighting Deficit

HIGH

The 128.9-kilometer road network is supported by 1,829 street lights. This equates to an average of **14.2 lights per kilometer** (spacing of 70.5 meters). Standard urban residential benchmarks require 30–40 lights per kilometer (spacing of 25–30 meters). The district is significantly under-lit, creating potential traffic and pedestrian safety issues.

Insight 5 · Street Lighting Deficit

14.2 lights/km vs 30–40 benchmark · spacing 70.5 m vs 25–30 m recommended

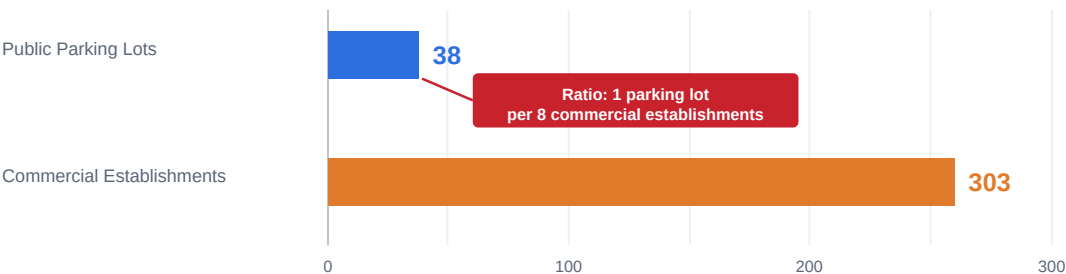


06 Public Parking Shortage in Commercial Zones

MEDIUM

The district boasts a vibrant commercial sector with **303 establishments**. However, there are only 38 designated public parking lots, resulting in a ratio of **0.13 public parking lots per commercial establishment**. This imbalance likely leads to congestion and illegal street parking during peak hours.

Insight 6 · Parking Shortage — 0.13 parking lots per commercial establishment

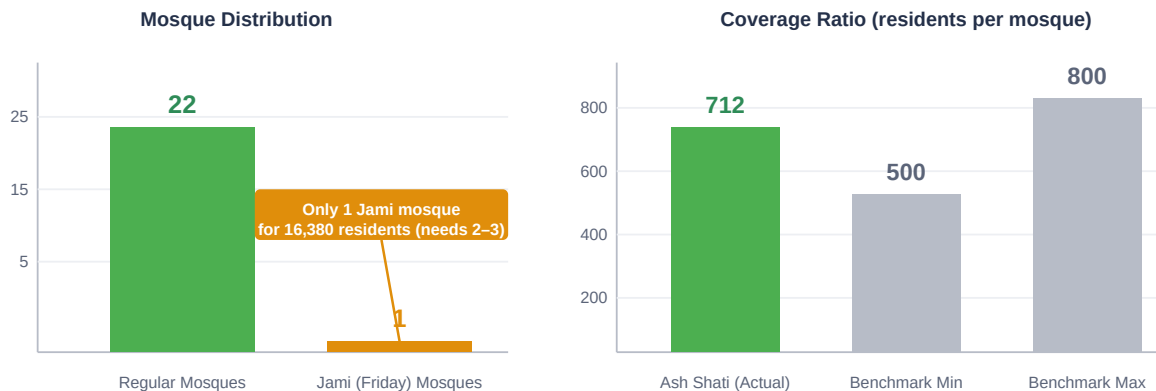


07 Mosque Coverage and Jami Shortage

MEDIUM

Overall mosque coverage is excellent, with **23 mosques** serving the population (1 mosque per 712 residents, perfectly within the Saudi benchmark of 1 per 500–800). However, there is only **1 Jami (Friday) mosque** registered for the entire district. A population of 16,380 typically requires 2–3 Jami mosques to accommodate Friday prayer capacity.

Insight 7 · Mosque Coverage — Adequate Overall, Jami Mosque Shortage

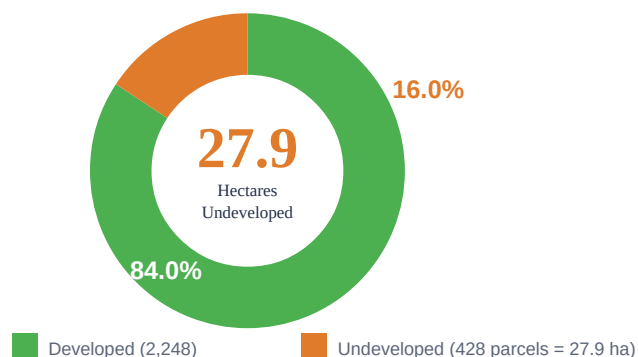


08 Undeveloped Land · Investment Opportunity

OPPORTUNITY

Despite being a mature coastal district, **428 parcels remain undeveloped**. This represents **27.9 hectares (279,201 m²)** of prime real estate. MOMAH can target these specific parcels for White Land Tax enforcement, or partner with developers to utilize this land to resolve the public school, healthcare, and green space deficits identified above.

Insight 8 · 27.9 Hectares of Prime Coastal Land Remain Undeveloped



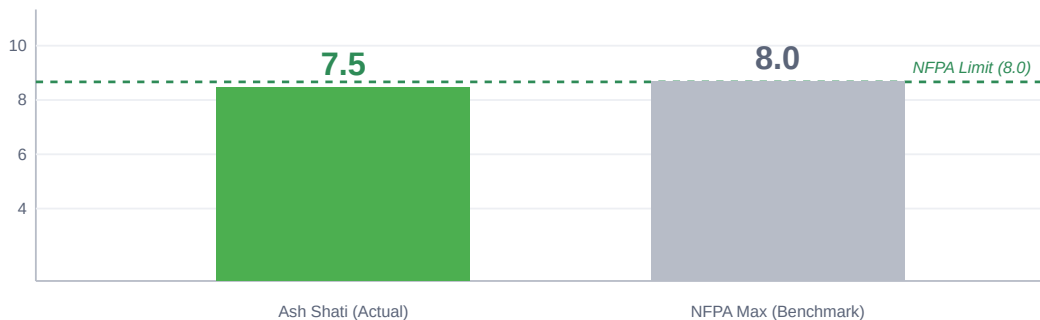
09 Fire Safety Preparedness

POSITIVE

The district performs exceptionally well in emergency preparedness. With **265 fire hydrants** serving 1,979 buildings, the ratio is **1 hydrant per 7.5 buildings**. This falls well within the strict NFPA standard benchmark of 1 hydrant per 5–8 buildings.

Insight 9 · Fire Safety — Within NFPA Standard

265 hydrants serving 1,979 buildings



10 Pedestrian Walkability

POSITIVE

The district features strong pedestrian infrastructure, with **591 dedicated pavement segments** covering **262,277 square meters**. This equates to an average of **2,034 m² of pavement per kilometer** of road, supporting the high commercial density and promoting a walkable urban environment.

Insight 10 · Pedestrian Walkability — Strong Infrastructure

591 pavement segments across 128.9 km road network



Strategic Recommendations

Six priority actions to translate findings into measurable outcomes.

1 • Activate Undeveloped Land for Public Goods

Direct the 27.9 ha of undeveloped parcels toward resolving the public school, healthcare, and green space deficits — combining White Land Tax enforcement with PPP development models.

2 • Launch Building Regularization Campaign

Address the 778 unlicensed parcels (29.1%) through a structured regularization initiative — generating municipal revenue while restoring regulatory compliance.

3 • Establish Primary Healthcare Network

Deploy at least 2 primary care centers to meet the WHO 1:10,000 benchmark, leveraging existing hospital-zoned parcels and undeveloped land.

4 • Upgrade Street Lighting & Public Safety

Increase lighting density from 14.2 to a minimum of 30 lights/km — closing a 53% gap and improving traffic and pedestrian safety across 128.9 km of roadway.

5 • Expand Friday Prayer Capacity

Designate 1-2 additional Jami mosques to align with population needs (16,380 residents typically require 2-3 Jami facilities).

6 • Resolve Commercial Parking Imbalance

Plan additional public parking infrastructure to support 303 commercial establishments — current ratio of 0.13:1 drives congestion and informal street parking.